

Lesson 1.18: Project Launch: Feed a Family of Four on \$40

Unit: Unit 1, Kitchen Skills and Smart Eating (Nutrition and Wellness) **Topic:** 1.4 Plan, Shop, Cook: Meals on a Budget **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes **Room:** Either (classroom or FACS lab; no cooking in this lesson)

Standards

- NYS module line: NYS N&W 1. Culinary Management e) Plan a nutritious meal
- NYS module line: NYS N&W 3. Consumer Resources and Finance a) Create and organize a shopping list
- NYS module line: NYS N&W 3. Consumer Resources and Finance b) Describe and demonstrate food purchasing strategies that maximize the benefits of the food budget
- NYS module line: NYS N&W 3. Consumer Resources and Finance c) Identify monetary influences on one's ability to select and purchase nutritious foods
- CTE theme line: CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses
- CTE theme line: CTE PSI 2. Design Process (Proactive) a) c. Defining constraints that must be adhered to
- CTE theme line: CTE PSI 1. Problem Solving e) Demonstrate personal development of problem-solving skills through practice of these skills in a variety of classroom applications
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students are able to budget their time and money.
- CDOS Standard 3a Universal Foundation Skills: Managing Resources
- National FCS 3.0: 14.3.1 Use current dietary guidelines to plan for nutrition and wellness needs.
- National FCS 3.0: 14.3.3 Show the ability to select, store, prepare, and serve food that is nourishing and looks good on the plate.

(Note on PSI: the unit plan called for a PSI 1. Problem Solving line on defining the problem or evaluating solutions. PSI 1 has no such line; its lines a) to d) are definitions and e) is practice. The “defining” and “constraints” lines live under PSI 2. Design Process (Proactive) a), so 2. a) c. is cited verbatim and 1. e) is cited for the practice. Four module lines are cited because this is the project launch and all four are assessed on the rubric.)

Enduring understanding and essential question

Big idea: A budget is a constraint, and a constraint is what turns a wish into a plan. Forty dollars for four people is enough for a good dinner if you plan the plate, use the circular,

and do the math to the penny. **Essential question:** Can \$40 feed a family of four a dinner they would actually want to eat?

Objectives

By the end of this lesson, students will be able to:

1. (Low) Given the project brief, describe the five deliverables (the priced plan, the MyPlate check, the cooked dish, the menu card, the pitch) and name each team role's job.
2. (Mid) Given a family card and this week's circular, plan one dinner for four that follows every rule on the card, covers at least four MyPlate groups, and is priced to the penny with a unit on every line.
3. (High) Given a first total that is over \$40 or a plate that is short a group, revise the plan (swap, downsize, or drop an item) and justify each change in one sentence.

Vocabulary

Tier 2 (general academic): brief, deliverable, constraint, revise, justify Tier 3 (FACS): budget cap, running total, serving, unit price, MyPlate check, station dish. See the unit vocabulary list for the Spanish column.

Materials and setup

- Project sheet, one per student: Project 01 - Feed a Family of Four on 40 Dollars.md
- Rubric, one per student: Rubric 01 - Feed a Family of Four.md
- Handout, one per team plus one per student to copy into: Handout 01.18 - Project Plan Sheet.md (planning grid, pricing table with a running total, MyPlate check, station dish box)
- Family cards from Lesson 1.14, one per team (Handout 01.14 - Meal Planning Grid and Family Cards.md, page 2)
- This week's circular, two per team
- Role cards (Planner, Shopper, Nutrition Checker, Presenter, plus the three expansion roles), one set per team
- Calculators, two per team
- The \$12 vs \$38 hook: two dinner photos and two receipts on the slides (see Slides 01.18)
- Slides: Slides 01.18 - Project Launch Feed a Family of Four on 40 Dollars (outline in 05 Materials; the site renders it as a deck)
- Setup notes: Teams are the lab station teams, about seven students each, three teams per class. Assign each team a different family card so no two teams in a class plan for the same family. Print the circular the day before and check that at least one protein, one grain, one vegetable, one fruit, and one dairy item are on sale, so a \$40 plan is reachable. Allergy check: no food is handled in this lesson; the allergy check for the cooked dish happens in Lesson 1.20 Day 1.

Pre-assessment

The exit cards from Lessons 1.14 to 1.17 and the clipboard checks tell you which teams can already price with a unit and which cannot. Seat one strong unit-price student on each team as the Shopper. The do now on Day 1 is the last read before the project starts.

Do now and hook (Time: 5 min, Day 1)

On the board: “Two families each made dinner for four last night. One spent \$12. One spent \$38. Which family ate better? Write one sentence and one reason.”

Debrief: hands for \$12, hands for \$38. Then the hook: show the two dinners. Dinner A, \$38.47: a name-brand frozen lasagna, a bagged salad kit, a bakery garlic bread, a two-liter soda, and a name-brand ice cream. Dinner B, \$12.16: store-brand pasta, a jar of sauce, frozen mixed vegetables, store-brand shredded parmesan, and a loaf of Italian bread. Ask the class to check both against MyPlate. Dinner B has grain, vegetable, dairy, and could add a protein for \$2. Dinner A has more money in it and no fruit. The point: the price and the plate are two different questions. “Your team gets \$40. Your job is to win both questions.”

Procedure

Step	Teacher will	Students will	Time
Day 1, 1. Direct instruction: the project brief	Walk through the project sheet: the scenario, the five deliverables, the calendar (plan today, price tomorrow, peer check, lab, pitch), and the rubric’s five criteria. Point at the two budgets: \$40 is the family’s dinner at home for four; \$20 is the lab budget to cook a class-size taste of one dish from that plan. Read the constraints aloud: no knives, pre-cooked meat only, stovetop and oven, teacher handles the oven and drains pasta.	Follow on the project sheet. Highlight the five deliverables. Ask questions.	8 min

Day 1, 2. Roles and the family card	Hand out role cards and family cards. Read each role's job in one sentence: the Planner runs the plan sheet and the plate; the Shopper owns the circular and the prices; the Nutrition Checker owns the MyPlate check and the family's rules; the Presenter owns the menu card and the pitch. Expansion roles for teams of five to seven: the Second Shopper prices the sides, the Recorder keeps the team copy clean, the Materials Manager runs calculators, circulars, and the running total. Teams choose roles in two minutes; if they cannot, the teacher assigns.	Choose roles, write names on the team plan sheet, read the family card aloud together, and underline every rule.	6 min
Day 1, 3. Learning activity: plan the dinner	Teams fill the planning grid: a main, a vegetable, a grain (if the main has none), a fruit or a second vegetable, a drink (water counts and is free), and one item the family already has in the pantry. Each dish gets a MyPlate group. The team picks the station dish: the one dish	Plan on the grid. Mark the MyPlate group for every dish. Circle the station dish. Each student copies the team plan onto their own sheet.	17 min

	from the plan they will cook in Lesson 1.20, which must fit the room's constraints. Circulate with the check questions.		
Day 1, closure	See Closure.	See Closure.	4 min
Day 2, do now	On the board: "Your plan has five dishes. Before you price anything, guess the total. Write it at the top of your sheet. You will find out how close you were."	Write a guess.	3 min
Day 2, 4. Pricing to the penny	Show the pricing table on the screen and price one item together: product, size, price, unit, quantity needed for four, cost, running total. Rules: every line has a unit; if a package is bigger than the family needs, the whole package is paid for (nobody sells half a jar); pantry items are \$0 and marked "pantry"; sale prices count only if the sale is on this week.	Shopper and Second Shopper find prices in the circular; Planner reads the quantity needed; Materials Manager runs the calculator; Recorder writes the running total.	16 min
Day 2, 5. MyPlate check and the revise loop	Nutrition Checker runs the MyPlate check on the sheet: which groups are covered, which are missing, and whether any rule on the family card is	Check the plate, revise, recalculate, log each change. Rough-cost the station dish against \$20 in the station dish box.	14 min

	broken. Then the revise loop: if the total is over \$40, or a group is missing, or a rule is broken, the team swaps, downsizes, or drops one item at a time, recalculates, and writes one sentence for each change in the revision log. Circulate; sign off on the sheet when it is at or under \$40 with four groups and no rule broken.		
Day 2, 6. Share the totals	Each team's Presenter says the guess, the first total, and the final total in one sentence. Write them on the board.	Report. Compare guesses to totals.	3 min
Day 2, closure	See Closure.	See Closure.	4 min

Day 1 totals 40 minutes: $5 + 8 + 6 + 17 + 4$. Day 2 totals 40 minutes: $3 + 16 + 14 + 3 + 4$.

Questions to ask

Monitor understanding:

- What is the station dish, and why does it have to be one of the dishes on your plan?
- Your family card says no peanuts. Where on your plan did you check for that?
- The sauce jar is 24 oz and the family needs 16 oz. What do you pay for?

Deepen learning:

- If you had to cut \$5 from this plan right now, what goes first and why? What stays no matter what?
- Which costs more per person, your dinner or the \$38 dinner from the hook? Which one would this family rather eat?
- What does a family do on a week when the sale items are not the things they eat?

Check for understanding

Day 1: clipboard yes or no per team on “every dish has a MyPlate group,” “station dish circled and it fits the constraints,” and “one pantry item marked.” Day 2: clipboard yes or no per team on “every line has a unit,” “running total matches the calculator,” and “revision log has a sentence per change.” A team missing the unit column gets the Shopper sent back to the circular, not a teacher fix.

Closure (Time: 4 min each day)

Day 1: exit card, 3 dishes on your team’s plan, 2 rules on your family card, 1 worry about the price. Two students read a worry aloud; write the worries on the board for tomorrow. Day 2: exit card, 3 items you priced and their unit prices, 2 changes your team made and why, 1 thing you want the peer check team to look at tomorrow. Three students read one line aloud. Sort into “got it” and “needs more”; a “needs more” on unit price gets the worked-example card in the project folder before Lesson 1.19.

Differentiation and supports

- ELL: the role cards have an icon and a one-sentence job; the family cards use short sentences with rules in bold; the plan sheet has the pricing table pre-labeled with a worked row; sentence starters in Spanish on the project sheet (“We chose ___ instead of ___ to save money,” “This family cannot eat , ” **“Our total is ”**); pair each ELL student with the Recorder so the team copy is a model to copy from.
- IEP and 504 (general): choice of role (the Materials Manager and Recorder roles have less writing and no public speaking); calculators for everyone; extended time by finishing the pricing at the start of Lesson 1.19; the visual plan template from the project sheet (boxes and pictures instead of a table); one team member’s copy can be a photo of the team sheet if handwriting is a barrier.
- Grade 6 support: the plan is four dishes, not five; the circular is pre-marked with the on-sale items in each MyPlate group; the pricing table has quantities for four already filled in by the teacher for the main.
- Grade 8 stretch: the team also prices the same dinner at name-brand prices and writes the difference; the station dish is costed to the penny (not rough-costed) with sales tax on the non-food items; the team writes a second version of the plan for a week when the family has \$30.
- UDL checkpoint used: multiple means of engagement (choice of dish within the family card, a real budget, a class vote at the end of the project) and multiple means of action and expression (roles let a student contribute through math, reading, drawing, or speaking).
- No-kitchen or no-machine alternative: not needed for this lesson; the no-kitchen version of the cooked dish is in Lesson 1.20 and is built into the station dish box (a no-cook dish is allowed).

Assessment

Objective	Standards met	Learning task	Assessment
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			(formative or summative; tool)
1. Describe the deliverables and the roles	PSI 2. a) c.; CDOS 3a Managing Resources	Brief and roles, Steps 1 and 2	Formative; Day 1 exit card; role names on the team sheet
2. Plan and price one dinner for four to the penny	N&W 1. e); N&W 3. a), b), c); FCL 4. c); Std 3 PI C; National FCS 14.3.1; National FCS 14.3.3	Plan, Step 3; pricing, Step 4	Summative; Rubric 01 criterion 1 (the plan and the math) and criterion 2 (nutrition and MyPlate), scored after the Lesson 1.19 revision
3. Revise the plan and justify each change	N&W 3. b); PSI 1. e); FCL 4. c)	Revise loop, Step 5	Formative today (revision log checked for one sentence per change); summative on Rubric 01 criterion 1

Homework

None. Optional: ask the person who shops at home what they would cook for four on \$40. Bring the idea; a team can use it.

Connections

Math: the running total and the unit column are the same skills the math teacher grades under ratios and decimals; share the plan sheet so the math class can check a team's arithmetic as a warm up. Music and PE: in my earlier version, music supplied the jingle and PE checked whether the meal fuels an active day. Both are optional partners for Lesson 1.21.

Sources

- NYS Middle Level CTE FACS Content Module 05, Nutrition and Wellness (nycetecenter.org, June 2018), lines 1. e), 3. a), 3. b), and 3. c). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 03, Financial and Consumer Literacy, line 4. c); Theme Module 05, Problem Solving and Innovation, lines 1. e) and 2. a) c. Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 3, Intermediate. Wording per 02 Standards/STANDARDS CROSSWALK.md, Part A.6.

- CDOS Intermediate Standard 3a, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 14.3.1 and 14.3.3, per 02 Standards/STANDARDS SOURCES - web research.md.
- My own materials: the Feed a Family of Four on \$40 run of show: the \$12 vs \$38 hook, the \$40 cap, team planning with flyers, and my own improvement note to add checkpoints during planning so no team reaches the pitch with a broken budget. The supermarket menu options and constraints.

Teacher notes

- If Day 1 runs short, cut the plan to four dishes and pick the station dish at the start of Day 2. If Day 2 runs short, skip Step 6 and let the totals be the Lesson 1.19 do now.
- My own note from the earlier version: add checkpoints during planning so no team reaches the jingle with a broken budget. That is why Day 2 has a teacher sign off and Lesson 1.19 is a whole day of checking. Do not skip the sign off.
- Common mistake: teams price the amount the recipe needs instead of the package they have to buy. Say the rule early: you pay for the jar, not the spoonful.
- Second common mistake: the station dish does not fit the constraints (a stir fry that needs knives, a raw chicken dish). Check the circled dish on Day 1 before anyone prices it. The three fallback dishes in Lesson 1.20 are there for a team whose dish cannot work.
- Water is a free drink and a real one. A team that “saves” \$3 by not buying soda has made a nutrition decision and a budget decision at once. Say so.